Logistic Regression and Classifications

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From Continuous to Categorical Outcome

$f($ $) \longrightarrow$ dog

$f($ $) \longrightarrow$ cat

- Response Y: discrete value
  - e.g., $Y = \{\text{dog, cat}\}$
  - or $Y = \{0, 1\}$, 1 - dog; 0 - not dog

Classification Methods

- K-Nearest Neighbor
- Logistic regression
- Classification tree
- Random forest
- Boosted tree
- Support vector machine
- Neural networks
- Deep learning
- ...
- Is clustering a classification model?

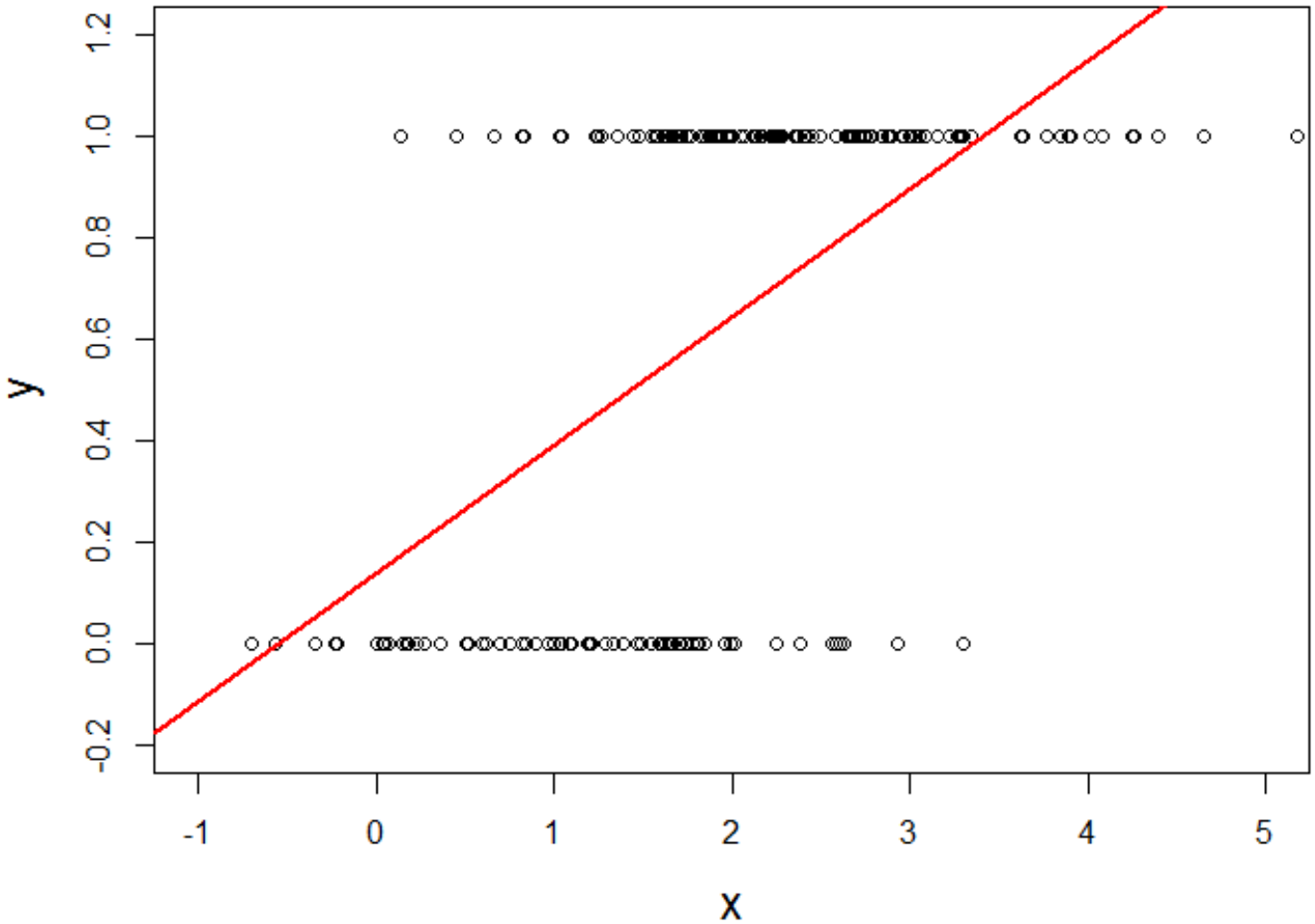
Why Not Linear Regression

- Example: default prediction
  - Default ($Y = 1$) vs. Nondefault ($Y = 0$)
  - X_1: credit card balance level, X_2: income level
- Suppose the estimated linear regression is

$$\hat{Y} = -1.5 + 2X_1 - X_2$$

- What is the predicted value if a person's balance level is 1 and income level is 3?
- How to interpret this value?

An Illustration



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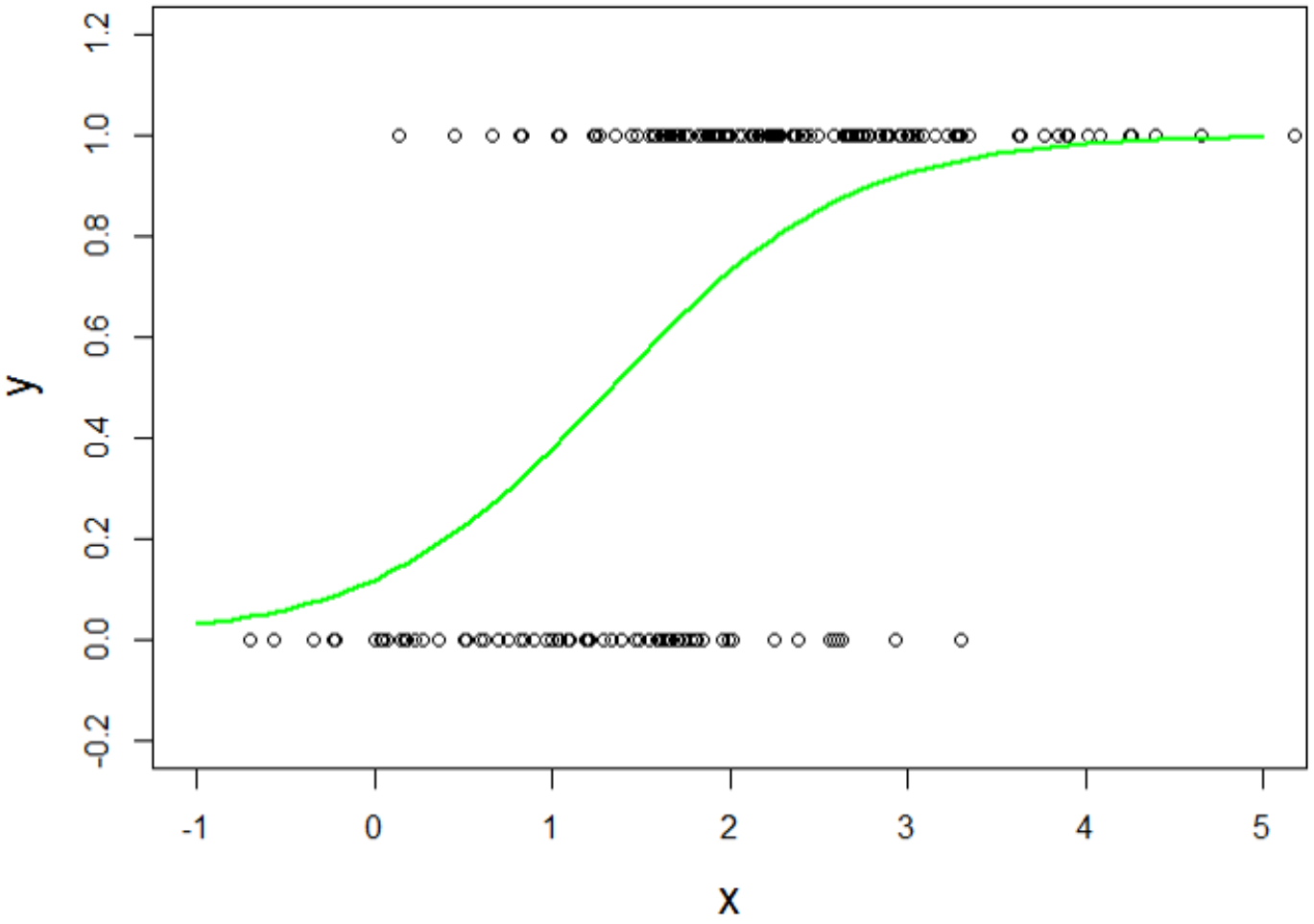


Classification

- Denote $\mathcal{C}(X)$ as a classifier
- Most DM algorithms produce probabilistic outcome
  - e.g. probability that X belongs to each class
- Classification is based on certain decision rules
- Example: The model prediction tells that the probability of default is 0.2, then

Threshold	<0.1	>0.1
Class	Nondefault	Default

Logistic Regression

- Generalized linear model
- Logistic model for binary response

$$\mathbb{P}(y_i = 1|\mathbf{x}_i) = \frac{1}{1 + \exp(-\beta_0 - \sum_{j=1}^{p} \beta_j x_{ij})}$$

- Sigmoid function
- Function of X
- Outcome is **predicted probability** of event
- More than two classes: *multinomial logistic model*

Loss Function (in machine learning)

- Recall LS

$$L_{OLS}(\beta) = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

- For logistic regression, we have a different loss

$$L_{logit}(\beta) = \sum_{i=1}^{n} -2 \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right]$$

where $p_i = \frac{1}{1+\exp(-\beta_0 - \sum_{j=1}^{p} \beta_j x_{ij})}$

Maximum Likelihood Estimation (in statistics)

- $y_i = \begin{cases} 1 & \text{with Prob. } p_i \\ 0 & \text{with Prob. } 1 - p_i \end{cases}$, p_i depends on $\mathbf{x}_i$ and unknown β
- Likelihood function of $y_i|\mathbf{x}_i$:

$$Likelihood_i = p_i^{y_i} \times (1 - p_i)^{1-y_i}$$

- log-likelihood for all observations:

$$logLik(\beta) = \sum_{i=1}^{n} [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$

- By simple algebra (show in exercise),

$$logLik(\beta) = \sum_{i=1}^{n} \left\{ y_i \mathbf{x}_i^T \beta - \log \left(1 + \exp(\mathbf{x}_i^T \beta) \right) \right\}$$

How to solve for β?

- No analytical solution like OLS
- Solve numerically (univariate example)
  - Gradient descent

$$\hat{\beta}^{(n+1)} = \hat{\beta}^{(n)} - \alpha * L'(\hat{\beta}^{(n)})$$

    where α is called learning rate.
  - Newton's method (a very good tutorial)

$$\hat{\beta}^{(n+1)} = \hat{\beta}^{(n)} - \frac{L'(\hat{\beta}^{(n)})}{L''(\hat{\beta}^{(n)})}$$

    - We call derivative for univariate: $f'(x)$, $f''(x)$
    - We call gradient for multivariate: $\nabla f(x)$, $\nabla^2 f(x)$

Odds and Interpretation of β

- Logistic model is also called log odds model
- Odds: ratio of probabilities:

$$\text{Odds}(X) = \mathbb{P}(Y = 1|X)/\mathbb{P}(Y = 0|X)$$

- Logit link (logit transformation)

$$\text{logit}(P) = \log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p$$

- By simple algebra, given all X's are fixed except X_j

$$\beta_j = \log\left(\frac{\text{Odds}(X_j + 1)}{\text{Odds}(X_j)}\right)$$

which is *log of odds ratio*.

Multinomial Logit Model

- Response $Y = 1, 2, ..., K$, K classes
- Given predictors $\mathbf{x}_i$

$$\log\left(\frac{\mathbb{P}(y_i = 2)}{\mathbb{P}(Y_i = 1)}\right) = \beta_2^T \mathbf{x}_i$$
$$\log\left(\frac{\mathbb{P}(y_i = 3)}{\mathbb{P}(Y_i = 1)}\right) = \beta_3^T \mathbf{x}_i$$
$$\vdots$$
$$\log\left(\frac{\mathbb{P}(y_i = K)}{\mathbb{P}(Y_i = 1)}\right) = \beta_K^T \mathbf{x}_i$$

- The first class "1" is the reference
- There are $(K-1) \times (p+1)$ coefficients need to be estimated.

Prediction — From Probability to Class

- Direct outcome of model: probability
- Next step: classification
- Need decision rule (cut-off probability – p-cut)
- Not unique

Confusion Matrix

- Classification table based on a specific cut-off probability
- Used for model assessment

	Pred=1	**Pred=0**
True=1	True Positive (TP)	False Negative (FN)
True=0	False Positive (FP)	True Negative (TN)

- FP: type I error; FN: type II error
- Different p-cut results in different confusion matrix
- Try to understand this table instead of memorizing!

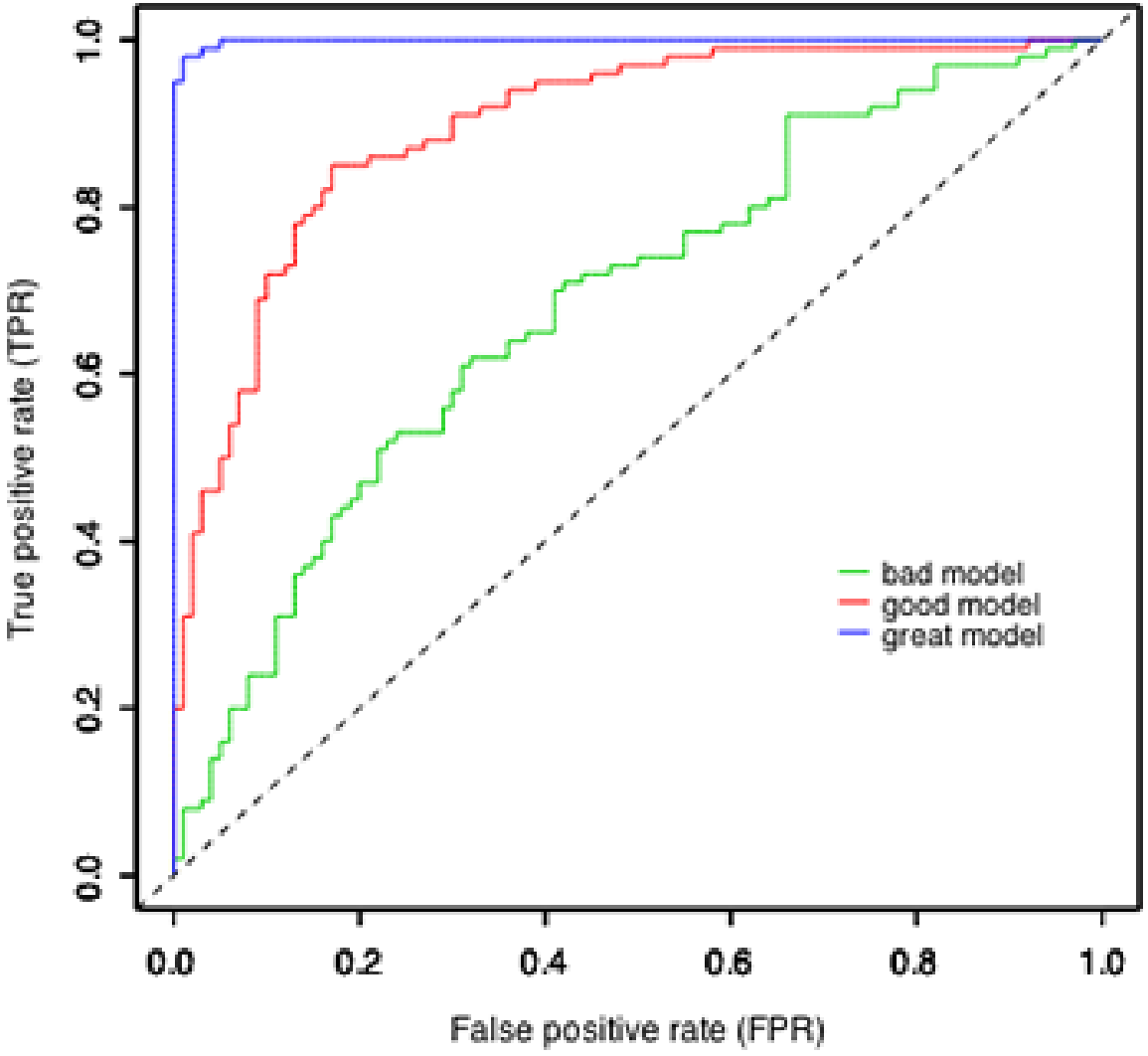
Some Useful Measures

- Misclassification rate (MR) $= \frac{\text{FP}+\text{FN}}{\text{Total}}$
- True positive rate (TPR) $= \frac{\text{TP}}{\text{TP}+\text{FN}}$: Sensitivity or Recall
- True negative rate (TNR) $= \frac{\text{TN}}{\text{FP}+\text{TN}}$: Specificity
- False positive rate (FPR) $= \frac{\text{FP}}{\text{FP}+\text{TN}}$: 1 – Specificity
- True negative rate (FNR) $= \frac{\text{FN}}{\text{TP}+\text{FN}}$: 1 – Sensitivity

ROC Curve

- Receiver Operating Characteristic
- Plot of FPR (X) against TPR (Y) at various p-cut values
- Overall model assessment (not for a particular decision rule)
- Unique for a given model
- Area under the curve (AUC): a measure of goodness-of-fit

ROC Curve


True positive rate (TPR)
False positive rate (FPR)
0.0
0.2
0.4
0.6
0.8
1.0
bad model
good model
great model

Asymmetric Cost

- Example: compare following two confusion matrices based on two p-cut values

	Pred=1	**Pred=0**
True=1	10	40
True=0	10	440

	Pred=1	**Pred=0**
True=1	40	10
True=0	130	320

- Which one is better? In terms of what?
- What if this is about loan application
  - $Y = 1$: default customer
  - Default will cost much more than reject a loan application

Choice of Decision Threshold (p-cut)

- Do NOT simply use 0.5!
- In general, we use grid search method to optimize a measure of classification accuracy/loss
  - Cost function (symmetric or asymmetric)
- Grid search with cross-validation